

ERG Analysis API v2.4.0

Tier 3 — T3-B: Normative Reference Traceability Matrix

Document ID	T3-B_Normative_Traceability_Matrix_v1_0
Project	ERG Analysis API — AI Fairness CIO (Charity Reg. No. 1218464)
Pipeline version	v2.4.0 (PIPELINE_VERSION = "2.4.0")
Normative JSON	data/normative_data_baker2025.json (schema_version 1.0)
Primary reference	Baker RA, Leo SM, Clowes WIN, et al. ISCEV standard full-field ERG reference limits from 407 healthy subjects. Doc Ophthalmol. 2025;150(2):47–64. DOI: 10.1007/s10633-025-10009-2
N subjects	407 healthy subjects across three age strata
Age strata	le35 (≤35y) 36to59 (36–59y) ge60 (≥60y)
Electrodes	Gold Foil (primary norms) DTL Fiber (Deming-corrected to Gold Foil equivalent)
OSF pre-registration	https://doi.org/10.17605/OSF.IO/6WA42
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Status	DRAFT — pending Hamid Tavakoli sign-off

1. Purpose

This document provides a complete audit trail from the Baker et al. (2025) published normative data through every transformation step to the final Z-score and traffic-light output on the clinical report. It is structured as a cross-walk: for every normative value held in the JSON file, the table records the exact JSON key path, the numerical value (mean and SD), the pipeline variable that consumes it, the report field where it appears, and a traceability status.

The document covers three normative data structures:

- Protocol normative ranges — amplitude and implicit time reference values for five ISCEV protocols, two electrode types, and three age strata (Section 4).
- DTL Deming regression transforms — Baker 2025 Table 2 coefficients used to convert DTL Fiber measurements to Gold Foil equivalent before Z-scoring (Section 5).
- b/a ratio reference — DA 3 only; not in the protocols block; sourced from a separate top-level JSON object (Section 6).

2. End-to-End Data Flow

The normative data flows through the pipeline in four stages:

- Stage 1 — Load. At ERGReportGenerator instantiation, `_find_norm_json()` locates `data/normative_data_baker2025.json` and `json.load()` reads it. `_to_tuple()` converts `{mu, sigma}` dicts to `(mu, sigma)` tuples. The `electrode_transform` block is read separately into `self.ELECTRODE_TRANSFORM`.
- Stage 2 — Select. `compute_z_scores()` calls `_age_stratum(age_years)` to resolve the Baker 2025 stratum key, then `_get_ref(electrode, protocol, stratum)` to retrieve the `(mu, sigma)` tuple for each parameter.
- Stage 3 — Transform (DTL only). For `dtl_fiber electrode`, `_apply_dtl_transform()` inverts the Baker 2025 Table 2 Deming regression: $GFE_{equivalent} = (STE_{measured} - intercept) / slope$. This converts the DTL measurement to its Gold Foil equivalent before Z-scoring against Gold Foil norms.
- Stage 4 — Score and classify. `calculate_z_score(value, mean, sd)` computes $Z = (value - mean) / sd$. `generate_traffic_light()` maps $\max|Z|$ to Green / Amber / Red using locked thresholds. Z-scores and traffic light appear in Layer 2 (clinical summary), Layer 4 (technical audit), FHIR Observation components, and the PDF metrics table.

3. Findings Summary

Result	Count
PASS	5
AMBER	3
FAIL	0
TOTAL	8

There are no FAIL findings. Three AMBERs are raised, all of low clinical impact:

- F1 — No schema version assertion at JSON load time. If the JSON file is replaced with a future incompatible schema, the pipeline will load silently and produce incorrect Z-scores.
- F4 — LA 3 a_amp Deming transform has a borderline $r^2 = 0.68$. This is documented in the JSON note field but is not surfaced as a Layer 4 flag in the clinical report.
- F6 — b/a ratio constants are hardcoded in the pipeline rather than loaded from the JSON `ba_ratio` block, making the JSON an incomplete single source of truth for normative values.

4. Protocol Normative Reference Cross-walk

Each row maps one (protocol, electrode, stratum, parameter) cell from the normative JSON to the pipeline variable and report field that consumes it. Null entries are shown in grey with status N/A — these are parameters not defined for that protocol per ISCEV 2022 (e.g. a-wave for DA 0.01).

Column guide: JSON Key Path = exact dot-notation path in normative_data_baker2025.json; μ = mean; σ = SD; Pipeline Variable = the features dict key and any transformation applied before Z-scoring; Report Field = the PDF metrics table row or Layer 4 dict key where the Z-score appears.

5. DTL Deming Regression Transform Cross-walk

Baker et al. (2025) Table 2 provides Deming regression coefficients for converting DTL Fiber (silver thread) amplitude measurements to their Gold Foil equivalent (STE scale). The pipeline inverts this transform: $GFE_equivalent = (STE_measured - intercept) / slope$, then Z-scores the GFE_equivalent against the Gold Foil normative values. Peak time parameters (a_imp, b_imp) are shown as N/A — no transform is applied as the paper confirms bias ≤ 1.6 ms.

Protocol	Parameter	JSON Key Path	Slope	Intercept	r ²	n	Method	Pipeline Use (inversion formula)	Status
DA 0.01	a_amp	electrode_transform.transform["DA 0.01"]["a_amp"]	null	null	—	—	—	Not applicable (a_amp not available for this protocol or peak time parameter)	N/A
	b_amp	electrode_transform.transform["DA 0.01"]["b_amp"]	0.46	24.38	0.73	46	Deming	measured → (val - intercept) / slope → Z against gold_foil mu/sigma	PASS

Protocol	Parameter	JSON Key Path	Slope	Intercept	r ²	n	Method	Pipeline Use (inversion formula)	Status
DA 3	a_amp	electrode_transform.transform["DA 3"]["a_amp"]	0.56	8.24	0.79	49	Deming	abs(measured) → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
	b_amp	electrode_transform.transform["DA 3"]["b_amp"]	0.48	36.75	0.73	47	Deming	measured → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
DA 10	a_amp	electrode_transform.transform["DA 10"]["a_amp"]	0.55	9.93	0.74	50	Deming	abs(measured) → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
	b_amp	electrode_transform.transform["DA 10"]["b_amp"]	0.48	37.92	0.72	49	Deming	measured → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
LA 3	a_amp	electrode_transform.transform["LA 3"]["a_amp"]	0.55	1.15	0.68	52	Deming	abs(measured) → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
	b_amp	electrode_transform.transform["LA 3"]["b_amp"]	0.65	-3	0.81	52	Deming	measured → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS
LA 30 Hz	a_amp	electrode_transform.transform["LA 30 Hz"]["a_amp"]	null	null	—	—	—	Not applicable (a_amp not available for this protocol or peak time parameter)	N/A
	b_amp	electrode_transform.transform["LA 30 Hz"]["b_amp"]	0.73	-9.54	0.85	52	Deming	measured → (val – intercept) / slope → Z against gold_foil mu/sigma	PASS

6. b/a Ratio Reference Cross-walk

The b/a ratio reference is stored as a top-level ba_ratio object in the normative JSON, separate from the protocols block. It applies to DA 3 protocol only. The normal range 1.8–3.5 is modelled as a Gaussian with mean = 2.65, SD = 0.425, so that ±2 SD ≈ [1.8, 3.5]. The pipeline currently hardcodes these constants rather than loading from the JSON (finding F6).

Item	Value / Reference	Status
JSON key path	ba_ratio (top-level object in normative_data_baker2025.json)	PASS
mean	ba_ratio.mean = 2.65	PASS
sigma	ba_ratio.sigma = 0.425	PASS
Normal range	1.8–3.5 (mean ± 2SD: 2.65 ± 0.850)	PASS
Protocol scope	DA 3 only (ba_ratio.protocol = "DA 3")	PASS
Pipeline variable	features["ba_ratio"] — computed as b_wave_raw / abs(a_wave_amplitude)	PASS
Z-score computation	ERGReportGenerator.compute_z_scores() line 1312–1317; uses BA_RATIO_MEAN=2.65, BA_RATIO_SD=0.425	PASS

Item	Value / Reference	Status
Report field	B/A Ratio — metrics table row 4 (erg_report_generator.py param_defs line 566)	PASS
DTL transform	NOT applied — ba_ratio is a dimensionless ratio; amplitude transforms cancel	PASS

7. Traceability Findings

#	Finding	Evidence	Status	Notes / Actions
F1	schema_version matches loader expectation	JSON schema_version = "1.0". Pipeline loader (_find_norm_json / json.load) reads file without schema version check. No version gate exists yet.	AMBER	ACTION: Add a schema version assertion at load time: assert _norm["schema_version"] == "1.0". Prevents silent loading of a future incompatible schema.
F2	electrode_note confirms DTL shares Gold Foil norms	electrode_note field documents that DTL fiber uses Gold Foil normative values with Deming correction applied at Z-score time. Pipeline code confirms this: ZSCORE_SUPPORTED_ELECTRODES includes dtl_fiber; _apply_dtl_transform() inverts Baker 2025 Table 2.	PASS	Traceability complete. Note field provides audit trail for the design decision.
F3	null_convention documented and honoured	null_convention = "null indicates parameter not applicable for this protocol". Pipeline _to_tuple() returns None for null values (line 1102–1105). compute_z_scores() guards with ref.get(param) is not None before computing Z.	PASS	Null handling is defensive and consistent throughout.
F4	LA 3 a_amp Deming r^2 = 0.68 — borderline flag	JSON note field: "Borderline r^2 =0.68 < 0.70; included per Baker et al. given interdependence of ERG components". Pipeline uses this transform without surfacing the borderline r^2 to the clinician.	AMBER	ACTION: Surface the borderline r^2 as a Layer 4 flag (e.g. dtl_la3_a_amp_borderline_r2 = True) so the clinician can see it in the technical audit section of the report. Low clinical impact but important for regulatory transparency.
F5	Peak time parameters: no Deming transform applied	electrode_transform.applies_to states amplitude parameters only. a_imp and b_imp entries in transform block are all null. Pipeline confirms: _apply_dtl_transform() returns raw value unchanged when param_transform is None.	PASS	Correct per Baker 2025: peak time bias ≤ 1.6 ms across all protocols; no transform needed.
F6	b/a ratio: not in normative JSON protocols block	ba_ratio is a top-level JSON object, separate from the protocols block. Pipeline reads it as hardcoded constants (BA_RATIO_MEAN, BA_RATIO_SD) rather than loading from JSON. Values match JSON (2.65, 0.425).	AMBER	ACTION: Refactor pipeline to load BA_RATIO_MEAN and BA_RATIO_SD from the JSON ba_ratio block at startup, making the normative JSON the single source of truth for all reference values.

#	Finding	Evidence	Status	Notes / Actions
F7	N=407 and DOI recorded in JSON metadata	n_subjects = 407; doi = "10.1007/s10633-025-10009-2". Written to Layer 3 normative_meta reference_source string in pipeline. Displayed in PDF subtitle.	PASS	Full traceability from report to primary source.
F8	Age strata keys match pipeline_age_stratum() mapping	JSON age_strata = ["le35","36to59","ge60"]. Pipeline _age_stratum() maps: ≤35 → "le35", 36–59 → "36to59", ≥60 → "ge60". String inputs "≤35y","36-59y","≥60y" also handled.	PASS	Key mapping is exact. No mismatch between JSON keys and pipeline lookup.

8. Remediation Actions

Three actions are required before T3-B sign-off. All are low risk and confined to small code changes.

#	Priority	Action	Owner	Target
F1	MEDIUM	Add schema version assertion in ERGReportGenerator.__init__() immediately after json.load(): assert _norm.get("schema_version") == "1.0", f"Unexpected normative JSON schema version: {_norm.get('schema_version')}"	Hamid Tavakoli	T3-B v1.1
F4	LOW	After DTL transform for LA 3 a_amp, set Layer 4 flag: if protocol == "LA 3" and param == "a_amp" and electrode == "dtl_fiber": z_scores["_la3_a_amp_dtl_r2_borderline"] = True. Add note to PDF Layer 4 section.	Hamid Tavakoli	T3-B v1.1
F6	LOW	In ERGReportGenerator.__init__(), after loading normative JSON, read: self.BA_RATIO_MEAN = _norm["_ba_ratio"]["mean"]; self.BA_RATIO_SD = _norm["_ba_ratio"]["sigma"]. Remove hardcoded values.	Hamid Tavakoli	T3-B v1.1

9. Sign-off

This document is classified DRAFT. It will be reclassified APPROVED upon completion of remediation items F1, F4, and F6 and counter-signature by Hamid Tavakoli.

Role	Name	Date
T3-B Auditor (Claude AI)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CIO (Approval)	Hamid Tavakoli	Pending

End of T3-B Normative Reference Traceability Matrix v1.0 — AI Fairness CIO (Reg. No. 1218464)